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TERM PROJECT - FINAL REPORT

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DUE DATE : 23/04/2026

COURSE ID & : ELK 453E Industrial Applications of
NAME Power Electronics

COURSE COOR. : Prof. Dr. S. Barış Öztürk

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1 INTRODUCTION

Non-linear loads such as diode-bridge rectifiers with bulk capacitive filters draw highly distorted, pulsed input currents that degrade the grid power factor (PF) and inject a significant harmonic spectrum into the line. To comply with harmonic regulations (e.g. IEC 61000-3-2) and to improve the utilisation of the utility supply, a Power Factor Correction (PFC) stage is commonly placed between the diode bridge and the DC bus of switching power supplies [1, 2].

Among the several candidate topologies, the *boost PFC* is the de-facto industry standard for universal-input, single-phase, medium-power applications because its input inductor is in series with the source, producing inherently continuous input current and simplifying EMI filter design [3]. When paired with the *average current-mode* control scheme, the inductor current is forced to track a scaled, rectified sinusoidal reference generated from the line voltage, and the output DC voltage is regulated by an outer low-bandwidth loop. This dual-loop cascaded architecture decouples the fast current dynamics from the slow voltage dynamics and allows each loop to be tuned independently.

In this report, the design, analysis and open-loop verification of a 500 W, 400 V DC, 50 kHz single-phase boost PFC converter that draws its input from a 240 V_{rms}/50 Hz mains are carried out. The control structure follows the per-unit, feed-forward-based average current-mode scheme described in the Texas Instruments application report SPRA902A [4]. The final report documents the design flow from the passive-component selection to closed-loop simulation, frequency-domain verification, and PF/THD evaluation:

- Part (a): system-level block diagram of the converter, selection of the boost inductor L and output capacitor C_o , derivation of the duty-ratio range, and voltage/current ratings of the MOSFET, boost diode and diode-bridge rectifier.
- Part (b): derivation of the switched-inductor state equations, their moving-average (large-signal) form, and the open-loop Simulink simulation of the averaged model showing input voltage/current and output voltage/inductor current waveforms.
- Part (c): small-signal linearisation around the large-signal operating trajectory, extraction of the inner current-loop plant $G_{id}(s)$ and the outer voltage-loop plant $G_{vi}(s)$, and Bode analysis at the target crossover frequencies.
- Parts (d)–(g): PI implementation, loop-gain stability margins, closed-loop waveforms, power factor, THD, and Model-in-the-Loop verification.

Design specifications of the converter are summarised in Table 1. All derivations use only the specifications of Table 1; no efficiency losses are considered at the design stage (i.e. $P_{in} = P_o$, $\eta = 1$), so that all ratings derived below are conservative with respect to the ideal operating point.

Table 1: Design specifications of the single-phase boost PFC converter.

Symbol	Description	Value
V_{in}	RMS line voltage	240 V _{rms}
f_{line}	Line frequency	50 Hz
V_o	Regulated DC output voltage	400 V
P_o	Rated output power	500 W
f_s	Switching frequency	50 kHz
$\Delta V_o/V_o$	Maximum allowable output voltage ripple	< 1 %
$\hat{V}_p$	PWM sawtooth carrier peak	3 V

2 PART (a): SYSTEM BLOCK DIAGRAM AND COMPONENT RATINGS

This section develops the topological block diagram of the overall converter, derives the operating-point quantities (input peak values, duty ratio range) and computes the passive and semiconductor component values that satisfy the Continuous-Conduction-Mode (CCM) constraint and the 1 % output-ripple specification. All derivations use only the specifications in Table 1 and assume a lossless converter ($\eta = 1$) at the design point.

2.1 Overall system topology

The complete block diagram is shown in Figure 1. It consists of:

- An EMI filter followed by a full-wave diode-bridge rectifier (DBR) that produces the rectified waveform $v_{rect}(t) = V_{in,pk} |\sin(\omega_{line}t)|$.
- A boost stage with inductor L , MOSFET S , ultrafast diode D and bulk output capacitor C_o feeding the resistive load $R = V_o^2/P_o$.
- A dual-loop average current-mode controller: an inner PI compensator $G_{ci}(s)$ regulating i_L , an outer PI compensator $G_{cv}(s)$ regulating V_o , a multiplier/divider block that generates the current reference $i_{ref} = (v_{ea} \cdot v_{rect})/V_{rms}^2$, and a PWM modulator with sawtooth peak $\hat{V}_p = 3$ V [4].

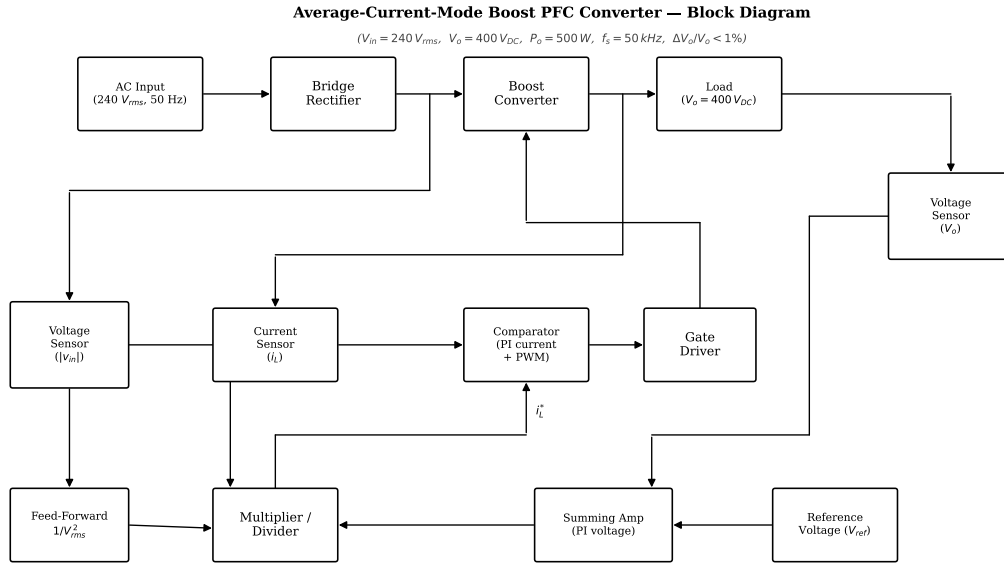


Figure 1: Overall block diagram of the single-phase average current-mode boost PFC converter with TI SPRA902A per-unit feed-forward control structure.

2.2 Input and output operating-point quantities

The rectifier output follows the rectified mains:

$$V_{in,pk} = \sqrt{2} V_{in} = \sqrt{2} \cdot 240 = 339.41 \text{ V.} \quad (1)$$

Under the unity-power-factor assumption, the fundamental of the line current is in phase with $v_{in}(t)$ and its RMS value follows directly from the active-power balance:

$$I_{in} = \frac{P_o}{V_{in}} = \frac{500}{240} = 2.0833 \text{ A}_{rms}, \quad (2)$$

$$I_{in,pk} = \sqrt{2} I_{in} = 2.9463 \text{ A.} \quad (3)$$

The average load current is

$$I_o = \frac{P_o}{V_o} = \frac{500}{400} = 1.25 \text{ A}, \quad R = \frac{V_o^2}{P_o} = \frac{400^2}{500} = 320 \Omega. \quad (4)$$

2.3 Duty ratio range

For a boost converter in CCM, the steady-state conversion ratio relates the instantaneous rectified input voltage to V_o :

$$\frac{v_{rect}(t)}{V_o} = 1 - d(t) \quad \implies \quad d(t) = 1 - \frac{V_{in,pk} |\sin(\omega_{line} t)|}{V_o}. \quad (5)$$

The instantaneous duty thus varies through the line cycle. Its bounds are obtained from Equation (5) at the two extreme instants of the rectified waveform:

- At the peak of the line voltage, $|\sin| = 1$, and the duty reaches its minimum

$$d_{min} = 1 - \frac{V_{in,pk}}{V_o} = 1 - \frac{339.41}{400} = 0.1515. \quad (6)$$

- Around the zero crossings of the line, $|\sin| \rightarrow 0$, so $d(t) \rightarrow 1$. In a practical implementation the PWM modulator is saturated (e.g. $d_{max} \leq 0.95$) to preserve a minimum off-time for the power MOSFET.

Hence the theoretical duty range is $0.1515 \leq d(t) < 1$, and the limited range used in the controller is $0.1515 \leq d(t) \leq 0.95$.

2.4 Boost inductor selection

During the switch on-time $t_{on} = dT_s$, the inductor is connected across $v_{rect}(t)$, so the peak-to-peak current ripple produced in one switching period is

$$\Delta i_L(t) = \frac{v_{rect}(t) d(t)}{L f_s}. \quad (7)$$

Substituting Equation (5) gives

$$\Delta i_L(t) = \frac{v_{rect}(t) [V_o - v_{rect}(t)]}{L V_o f_s}. \quad (8)$$

A standard CCM design rule fixes the peak-to-peak ripple at the top of the line cycle to a fraction k of the peak fundamental current, typically $k = 0.2$ – 0.4 . Evaluating Equation (8) at $v_{rect} = V_{in,pk}$ and imposing $\Delta i_{L,pk} = k I_{in,pk}$:

$$L \geq \frac{V_{in,pk} (V_o - V_{in,pk})}{k I_{in,pk} V_o f_s}. \quad (9)$$

With $k = 0.4$,

$$L \geq \frac{339.41 \cdot (400 - 339.41)}{0.4 \cdot 2.9463 \cdot 400 \cdot 50 \times 10^3} = \frac{20\,568}{23.57 \times 10^6} = 0.872 \text{ mH}. \quad (10)$$

Selecting $L = 12 \text{ mH}$ provides a large margin over the CCM limit and yields the actual peak current ripple

$$\Delta i_{L,pk} = \frac{V_{in,pk} (V_o - V_{in,pk})}{L V_o f_s} = \frac{20\,568}{12 \times 10^{-3} \cdot 400 \cdot 50 \times 10^3} = 0.0857 \text{ A}, \quad (11)$$

i.e. approximately 2.9% of $I_{in,pk}$. The larger inductance reduces the switching-frequency ripple considerably and keeps the converter robustly inside CCM over the line cycle except for the unavoidable narrow interval around the zero crossing in a single-stage boost PFC.

2.5 Output capacitor selection

Because the input power of a single-phase PFC follows

$$p_{in}(t) = v_{in}(t) i_{in}(t) = V_{in,pk} \sin(\omega_{line} t) \cdot I_{in,pk} \sin(\omega_{line} t) = P_o [1 - \cos(2\omega_{line} t)], \quad (12)$$

the output capacitor C_o must absorb the second-harmonic AC component $-P_o \cos(2\omega_{line} t)$. Neglecting the switching-frequency ripple (which is filtered by L and C_o), the low-frequency capacitor current is

$$i_{C,ac}(t) = \frac{-P_o \cos(2\omega_{line} t)}{V_o}. \quad (13)$$

Integration yields the $2\omega_{line}$ voltage ripple,

$$v_{C,ripple}(t) = \frac{1}{C_o} \int i_{C,ac} dt = -\frac{P_o}{2\omega_{line}V_oC_o} \sin(2\omega_{line}t), \quad (14)$$

so that the peak-to-peak ripple becomes

$$\Delta V_{o,pp} = \frac{P_o}{\omega_{line}V_oC_o} = \frac{P_o}{2\pi f_{line}V_oC_o}. \quad (15)$$

Enforcing the 1 % specification, $\Delta V_{o,pp} \leq 0.01 V_o = 4 \text{ V}$, gives the minimum capacitance:

$$C_o \geq \frac{P_o}{2\pi f_{line}V_o(0.01 V_o)} = \frac{500}{2\pi \cdot 50 \cdot 400 \cdot 4} = 994.7 \text{ }\mu\text{F}. \quad (16)$$

To satisfy the ripple requirement while avoiding an unnecessarily slow output-voltage dynamic, the final design value is chosen as $C_o = 1500 \text{ }\mu\text{F}$, giving

$$\Delta V_{o,pp} = \frac{500}{2\pi \cdot 50 \cdot 400 \cdot 1500 \times 10^{-6}} = 2.65 \text{ V} \implies \frac{\Delta V_o}{V_o} = 0.66 \% < 1 \%. \quad (17)$$

2.6 Semiconductor ratings

Boost MOSFET. When S is off, $v_{DS} = V_o = 400 \text{ V}$ (plus switching-ringing overshoot). The peak switch current equals the peak inductor current,

$$I_{S,pk} = I_{in,pk} + \frac{1}{2}\Delta i_{L,pk} = 2.946 + 0.343 = 3.29 \text{ A}, \quad (18)$$

and the RMS switch current is bounded above by $I_{in,pk}\sqrt{D_{eff}} \leq 1.5 \text{ A}$ because $D(t) \leq 0.15$ at the top of the line cycle. Allowing a 25 % voltage safety margin and a $2\times-3\times$ current derating for thermal design, a 600 V/10 A super-junction MOSFET (e.g. IPW60R125P6 or STP10NK60) is chosen.

Boost diode. The boost diode is reverse-biased by $V_o = 400 \text{ V}$ during t_{on} and conducts the rectified output current in t_{off} . Its ratings follow from

$$V_{D,rev} = V_o = 400 \text{ V}, \quad I_{D,avg} = I_o = 1.25 \text{ A}, \quad I_{D,pk} \approx I_{in,pk} = 2.95 \text{ A}. \quad (19)$$

A 600 V, 5 A ultrafast-recovery ($t_{rr} < 50 \text{ ns}$) or SiC Schottky diode is selected (e.g. STTH8S06 or C3D04060).

Diode-bridge rectifier. Each bridge diode blocks a maximum reverse voltage of $V_{in,pk} = 339 \text{ V}$, carries an average current of $2I_{in}/\pi = 1.33 \text{ A}$ (half-cycle average), an RMS current of $I_{in}/\sqrt{2} = 1.47 \text{ A}$, and a peak current of $I_{in,pk} = 2.95 \text{ A}$. A 600 V, 6 A bridge (e.g. GBPC606 or KBU6K) with adequate thermal performance is adequate.

All selected components are summarised in Table 2.

Table 2: Summary of component values and semiconductor ratings.

Component	Parameter	Derivation	Value
Boost inductor L	40% current ripple	Eq. (9)	12 mH
Output capacitor C_o	<1% output ripple	Eq. (16)	1500 μ F
Load resistor R	V_o^2/P_o	Eq. (4)	320 Ω
MOSFET S	$V_{DS}, I_{D,pk}$	Eq. (18)	600 V/10 A
Boost diode D	$V_R, I_{F,pk}$	Eq. (19)	600 V/5 A
Bridge rectifier	$V_R, I_{F,avg}$	–	600 V/6 A
Duty-ratio range	$d_{min} - d_{max}$	Eq. (6)	0.1515 – 0.95

3 PART (b): AVERAGE MODEL AND OPEN-LOOP SIMULATION

In this section, the large-signal averaged model of the boost PFC converter is developed from its switched state equations and then simulated in open loop using MATLAB[®]/Simulink[®] Simscape Electrical[™]. The averaging step eliminates the switching-frequency components and yields a continuous-time nonlinear model that is the starting point for the small-signal analysis of part (c).

3.1 Switched state equations

Let $i_L(t)$ denote the instantaneous inductor current and $v_o(t)$ the instantaneous output capacitor voltage. Defining the switching function

$$q(t) = \begin{cases} 1 & \text{MOSFET ON } (0 \leq t < dT_s), \\ 0 & \text{MOSFET OFF } (dT_s \leq t < T_s), \end{cases} \quad (20)$$

Kirchhoff's voltage and current laws applied to the boost stage give two topological subintervals:

Subinterval I – MOSFET ON ($q = 1$):

$$L \frac{di_L}{dt} = v_{rect}(t), \quad (21)$$

$$C_o \frac{dv_o}{dt} = -\frac{v_o}{R}. \quad (22)$$

Subinterval II – MOSFET OFF ($q = 0$):

$$L \frac{di_L}{dt} = v_{rect}(t) - v_o, \quad (23)$$

$$C_o \frac{dv_o}{dt} = i_L - \frac{v_o}{R}. \quad (24)$$

Combining the two subintervals through the switching function yields the exact switched model

$$L \frac{di_L}{dt} = v_{rect}(t) - [1 - q(t)]v_o(t), \quad (25)$$

$$C_o \frac{dv_o}{dt} = [1 - q(t)]i_L(t) - \frac{v_o(t)}{R}. \quad (26)$$

3.2 Moving-average (large-signal) model

Applying the moving-average operator over one switching period

$$\langle x(t) \rangle_{T_s} \triangleq \frac{1}{T_s} \int_{t-T_s}^t x(\tau) d\tau, \quad (27)$$

and noting that $\langle q(t) \rangle_{T_s} = d(t)$, the state equations (25)–(26) become

$$L \frac{d\langle i_L \rangle}{dt} = \langle v_{rect} \rangle - [1 - d(t)]\langle v_o \rangle, \quad (28)$$

$$C_o \frac{d\langle v_o \rangle}{dt} = [1 - d(t)]\langle i_L \rangle - \frac{\langle v_o \rangle}{R}. \quad (29)$$

Because $f_s = 50 \text{ kHz} \gg 2f_{line} = 100 \text{ Hz}$, the line-frequency components of i_L , v_{rect} and v_o survive the averaging operator essentially unchanged; only the switching-frequency ripple is filtered. In state-space form, defining $\mathbf{x} = [\langle i_L \rangle, \langle v_o \rangle]^T$ and $u = \langle v_{rect} \rangle$:

$$\frac{d}{dt} \begin{bmatrix} \langle i_L \rangle \\ \langle v_o \rangle \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1-d}{L} \\ \frac{1-d}{C_o} & -\frac{1}{RC_o} \end{bmatrix} \begin{bmatrix} \langle i_L \rangle \\ \langle v_o \rangle \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u. \quad (30)$$

Equation (30) is *bilinear* in d and the states and is the continuous-time model implemented in the Simulink diagram used for the open-loop test.

3.3 Steady-state trajectory

Because the line voltage varies slowly compared to the switching dynamics, a quasi-steady-state (QSS) argument can be applied: setting $d\langle i_L \rangle/dt = 0$ in Equation (28) on a time scale longer than T_s but shorter than $T_{line}/2$ gives the DC conversion ratio

$$\langle v_o \rangle = \frac{\langle v_{rect} \rangle}{1 - d(t)}, \quad (31)$$

and unity-PF operation requires the averaged inductor current to follow the rectified input:

$$\langle i_L(t) \rangle = I_{in,pk} |\sin(\omega_{line} t)|. \quad (32)$$

Combining Equations (31) and (32) recovers the instantaneous duty of Equation (5), demonstrating internal consistency of the averaged model with the design of part (a).

3.4 Open-loop simulation

The averaged model of Equation (30) was implemented in Simulink using the final passive values $L = 12 \text{ mH}$ and $C_o = 1500 \text{ }\mu\text{F}$. A fixed duty command around the nominal operating point was used in the open-loop test. Since no feedback controller is active in this part, the simulation mainly verifies the power-stage behavior and the relation between the rectified input voltage, inductor current, and output capacitor voltage.

The selected steady-state DC-side waveforms are shown in Figure 2. The output voltage remains close to the expected 400 V operating region and the inductor current shows a rectified-line-frequency envelope with small switching ripple. With the final inductor value, the analytically predicted peak current ripple at the line-voltage peak is only $\Delta i_{L,pk} \approx 0.0857 \text{ A}$, which is much lower than the earlier lower-inductance design.

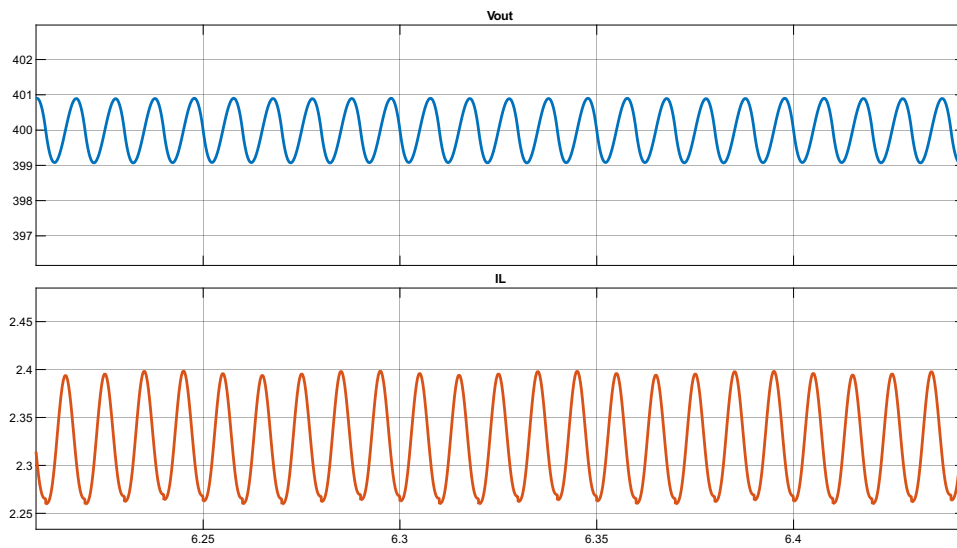


Figure 2: Open-loop DC-side waveforms: output voltage $v_o(t)$ and inductor current $i_L(t)$ using $L = 12 \text{ mH}$ and $C_o = 1500 \text{ }\mu\text{F}$.

The corresponding input-side waveforms are given in Figure 3. The mains voltage is sinusoidal and the input current follows the rectified PFC power-stage behavior. Because the loop is still open, this current is not yet optimally shaped for unity power factor; that objective is achieved only after closing the current and voltage loops.

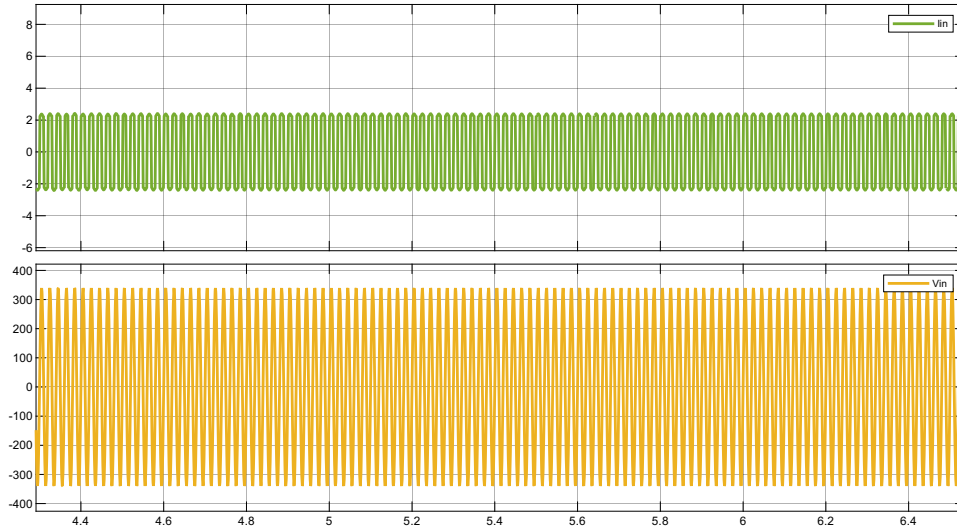


Figure 3: Open-loop input-side waveforms: input current $i_{in}(t)$ and input voltage $v_{in}(t)$.

Interpretation. The open-loop results confirm that the power stage operates in the expected boost-PFC region. The output capacitor keeps the DC-link voltage around the required level, while the input and inductor currents show the expected line-frequency envelope. These plots also show why closed-loop control is required: without the current-loop and voltage-loop PIs, the converter cannot guarantee a high power factor, low THD, or tight transient regulation. The closed-loop behavior is therefore evaluated in the following sections.

4 PART (c): SMALL-SIGNAL TRANSFER FUNCTIONS AND PLANT BODE PLOTS

The averaged model of Equations (28)–(29) is nonlinear because the duty ratio multiplies the state variables. For the controller analysis, the model is linearised around the rated operating point and the two plant transfer functions required by the cascaded PFC controller are obtained: the duty-to-inductor-current plant $G_{id}(s)$ and the inductor-current-to-output-voltage plant $G_{vi}(s)$. The final passive values used in this stage are

$$L = 12 \text{ mH}, \quad C_o = 1500 \text{ }\mu\text{F}, \quad R = 320 \text{ }\Omega, \quad V_o = 400 \text{ V}. \quad (33)$$

4.1 Small-signal linearisation

Each averaged variable is written as the sum of an operating-point value and a small perturbation,

$$i_L = I_L + \tilde{i}_L, \quad v_o = V_o + \tilde{v}_o, \quad d = D + \tilde{d}, \quad v_{rect} = V_{rect} + \tilde{v}_{rect}. \quad (34)$$

Substitution into the large-signal averaged equations and removal of the second-order perturbation products gives

$$L \frac{d\tilde{i}_L}{dt} = \tilde{v}_{rect} - (1 - D)\tilde{v}_o + V_o\tilde{d}, \quad (35)$$

$$C_o \frac{d\tilde{v}_o}{dt} = (1 - D)\tilde{i}_L - I_L\tilde{d} - \frac{\tilde{v}_o}{R}. \quad (36)$$

These equations show the two important control paths: the inner loop changes the inductor current through the duty-ratio perturbation, and the outer loop changes the output voltage by commanding the current reference.

4.2 Current-loop plant

For the implemented Bode analysis, the current-loop plant is written in the second-order form used in the MATLAB script `PFC_Bode_Analysis.m`:

$$G_{id}(s) = \frac{\tilde{i}_L(s)}{\tilde{d}(s)} = \frac{V_o(1 + sRC_o)}{s^2LRC_o + sL + R}. \quad (37)$$

Compared with the pure-integrator approximation $V_o/(sL)$, this expression keeps the output-capacitor dynamics and therefore captures the low-frequency resonance visible in the plant Bode plot. With $L = 12$ mH and $C_o = 1500$ μ F, the LC resonance is located at approximately

$$f_{LC} = \frac{1}{2\pi\sqrt{LC_o}} = 37.5 \text{ Hz}. \quad (38)$$

At the target current-loop crossover $f_{c,i}^{tgt} = 5$ kHz, the plant magnitude and phase are

$$|G_{id}(j2\pi f_{c,i}^{tgt})| = 0.515 \text{ dB}, \quad \angle G_{id}(j2\pi f_{c,i}^{tgt}) \approx -90^\circ. \quad (39)$$

4.3 Voltage-loop plant

Once the inner current loop is closed, the outer voltage loop sees the inductor-current command as the low-frequency control input. The plant used in the Bode analysis is

$$G_{vi}(s) = \frac{\tilde{v}_o(s)}{\tilde{i}_L(s)} = \frac{R}{1 + sRC_o}. \quad (40)$$

The output pole is therefore

$$f_p = \frac{1}{2\pi RC_o} = 0.332 \text{ Hz}. \quad (41)$$

At the target voltage-loop crossover $f_{c,v}^{tgt} = 2.5$ Hz, the plant response is

$$|G_{vi}(j2\pi f_{c,v}^{tgt})| = 32.48 \text{ dB}, \quad \angle G_{vi}(j2\pi f_{c,v}^{tgt}) \approx -82.45^\circ. \quad (42)$$

4.4 Plant Bode plots

Figure 4 shows the plant Bode diagrams generated using the final L and C_o values. The dashed vertical lines mark the nominal target crossovers of 5 kHz for the current loop and 2.5 Hz for the voltage loop.

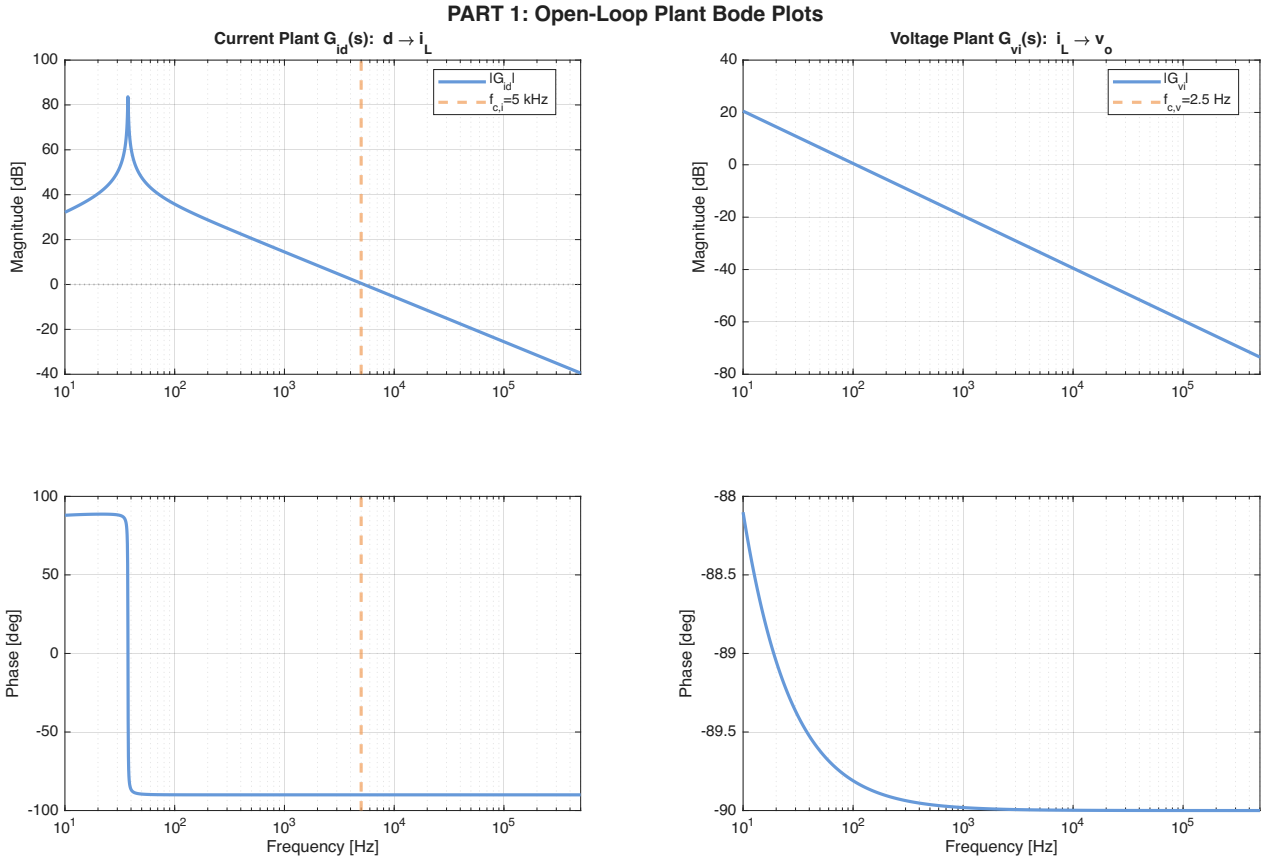


Figure 4: Open-loop plant Bode plots obtained with $L = 12$ mH and $C_o = 1500$ μ F. Left: current plant $G_{id}(s)$ from duty ratio to inductor current. Right: voltage plant $G_{vi}(s)$ from inductor current to output voltage.

The current plant has nearly -90° phase around the target crossover, while the voltage plant is already close to a single-pole response at 2.5 Hz. These two observations are important for interpreting the compensated loop margins in Section 6.

5 PART (d): PI CONTROLLER IMPLEMENTATION

This section gives the final PI controller values and the corresponding digital implementation constants used in the Simulink/PLECS model. In the initial design stage, the current-loop and voltage-loop crossover targets were selected as 5 kHz and 2.5 Hz, respectively. However, the PI values that moved the crossover frequencies closer to these targets did not produce an acceptable input-current harmonic performance; in repeated simulations, the THD could not be reduced below the 5% limit.

Therefore, the final design was obtained by testing several PI combinations together with different passive component values. The best practical result was achieved with

$$L = 12 \text{ mH}, \quad C_o = 1500 \text{ } \mu\text{F}. \quad (43)$$

Compared with the earlier passive selection, the boost inductor was increased from 1 mH to 12 mH, while the output capacitor was reduced from 4500 μ F to 1500 μ F. This passive redesign reduced the current ripple and improved the THD behavior more effectively than forcing the PI controller to satisfy only the nominal crossover-frequency targets.

5.1 Final continuous PI controllers

The controller blocks use the standard continuous PI form

$$G_c(s) = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s}. \quad (44)$$

The final selected controller values are

$$G_{ci}(s) = 2.8 + \frac{17600}{s}, \quad (45)$$

$$G_{cv}(s) = 0.055 + \frac{0.2}{s}. \quad (46)$$

Here, $G_{ci}(s)$ is the inner current-loop PI and $G_{cv}(s)$ is the outer voltage-loop PI. These gains should be interpreted as the best experimentally obtained set for the final switched simulation, not as the set that exactly matches the nominal crossover-frequency targets.

Table 3: Final continuous PI parameters used in the simulations.

Quantity	Current loop	Voltage loop
Proportional gain K_p	2.8	0.055
Integral gain K_i	17600	0.2
Nominal target crossover	5 kHz	2.5 Hz
Final passive values	$L = 12 \text{ mH}$, $C_o = 1500 \text{ } \mu\text{F}$	
Design selection criterion	Best observed THD/PF behavior after repeated trials	

5.2 Discrete implementation constants

For the discrete controller implementation, the simulation sampling time was

$$T_s = 20 \times 10^{-6} \text{ s} = 20 \text{ } \mu\text{s}. \quad (47)$$

The proportional gains were kept unchanged, while the integral constants were scaled by the sampling time:

$$K_{i,d} = K_i T_s. \quad (48)$$

Thus, the discrete values used in the model are

$$K_{p,v,d} = 0.055, \quad K_{i,v,d} = 0.2 \times 20 \times 10^{-6} = 4.0 \times 10^{-6}, \quad (49)$$

$$K_{p,i,d} = 2.8, \quad K_{i,i,d} = 17600 \times 20 \times 10^{-6} = 0.352. \quad (50)$$

Table 4: Continuous and discrete PI constants used in the final implementation.

Controller	K_p	K_i	$K_{p,d}$	$K_{i,d} = K_i T_s$
Voltage PI	0.055	0.2	0.055	4.0×10^{-6}
Current PI	2.8	17600	2.8	0.352

5.3 Closed-loop time-domain verification

The final PI values were first checked in time domain. Figure 5 shows the output-voltage response. The voltage rises from zero, exhibits a short startup transient, and settles close to the 400 V reference. The remaining ripple is mainly the expected double-line-frequency component of a single-phase PFC converter.

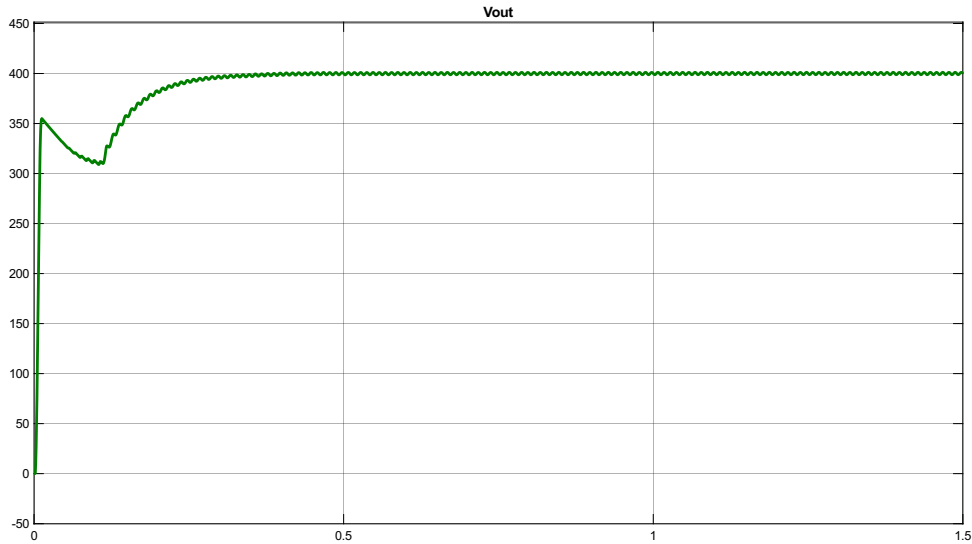


Figure 5: Closed-loop output voltage with the final selected PI values. The DC-link voltage settles around the 400 V reference.

Figure 6 shows the input-side voltage and current waveforms. The current waveform follows the line voltage envelope with switching ripple superimposed on it. This confirms that the current loop is actively shaping the input current.

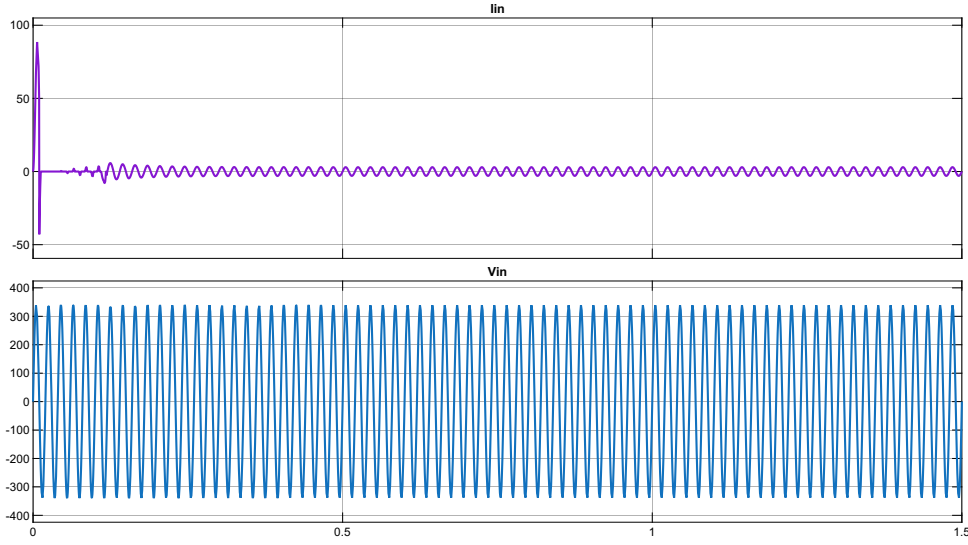


Figure 6: Closed-loop input current i_{in} and input voltage v_{in} obtained with the final PI implementation.

A magnified view of the inductor current is given in Figure 7. The envelope is periodic at the rectified line frequency and the switching ripple remains bounded, which is consistent with the selected inductor value.

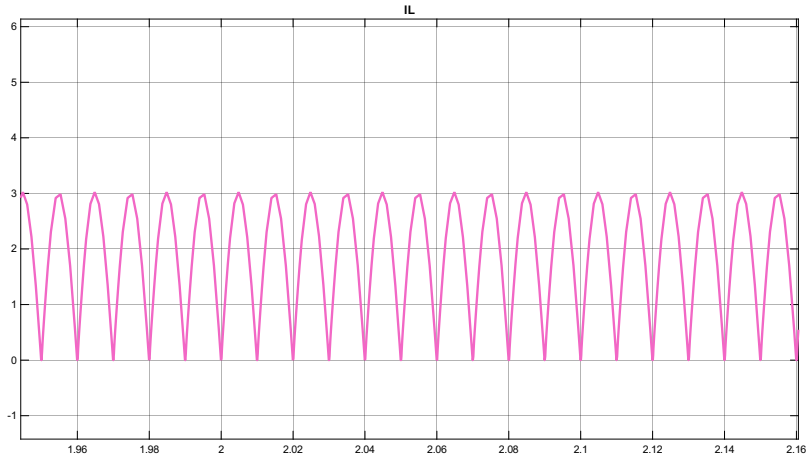


Figure 7: Zoomed inductor-current waveform obtained in the closed-loop simulation.

The time-domain results verify that the selected PI and passive values produce stable regulation. The exact frequency-domain bandwidth and margins of the same final controller are evaluated in the next section.

6 PART (e): LOOP-GAIN BODE PLOTS AND STABILITY MARGINS

This section evaluates the compensated current and voltage loops using the final passive values and the final PI gains of Section 5. The purpose of this analysis is to report the actual loop behavior of the implemented controller. Although the original crossover-frequency targets were 5 kHz for the current loop and 2.5 Hz for the voltage loop, the controller was finally selected according to the best observed time-domain, PF and THD performance. PI values that were closer to the nominal crossover targets were also tested, but they did not reduce the THD below 5 %.

6.1 Loop-gain expressions

The current open-loop transfer function is

$$L_i(s) = G_{ci}(s)G_{id}(s), \quad (51)$$

where $G_{ci}(s)$ is given in Equation (45) and $G_{id}(s)$ is given in Equation (37). The voltage open-loop transfer function includes the closed current-loop dynamics,

$$L_v(s) = G_{cv}(s)G_{vi}(s)T_i(s), \quad T_i(s) = \frac{L_i(s)}{1 + L_i(s)}. \quad (52)$$

The corresponding closed-loop transfer functions are

$$T_i(s) = \frac{L_i(s)}{1 + L_i(s)}, \quad T_v(s) = \frac{L_v(s)}{1 + L_v(s)}. \quad (53)$$

6.2 Open-loop margins

The open-loop Bode plots with stability margins are shown in Figure 8. The current-loop gain crossover is

$$f_{c,i} = 14.89 \text{ kHz}, \quad (54)$$

whereas the original target was 5 kHz. Thus, the actual current-loop crossover is approximately three times higher than the target. This result indicates that the final current proportional gain $K_{p,i} = 2.8$ gives a faster current loop than the nominal bandwidth target. However, lowering the gain only to match the target crossover did not give the best harmonic result in the switched model.

For the voltage loop, the measured gain crossover is

$$f_{c,v} = 5.85 \text{ Hz}, \quad (55)$$

while the nominal target was 2.5 Hz. This is approximately 2.3 times higher than the target. Nevertheless, it is still far below the 50 Hz line frequency, so the voltage loop remains slow enough for PFC operation and does not directly follow the line-frequency waveform.

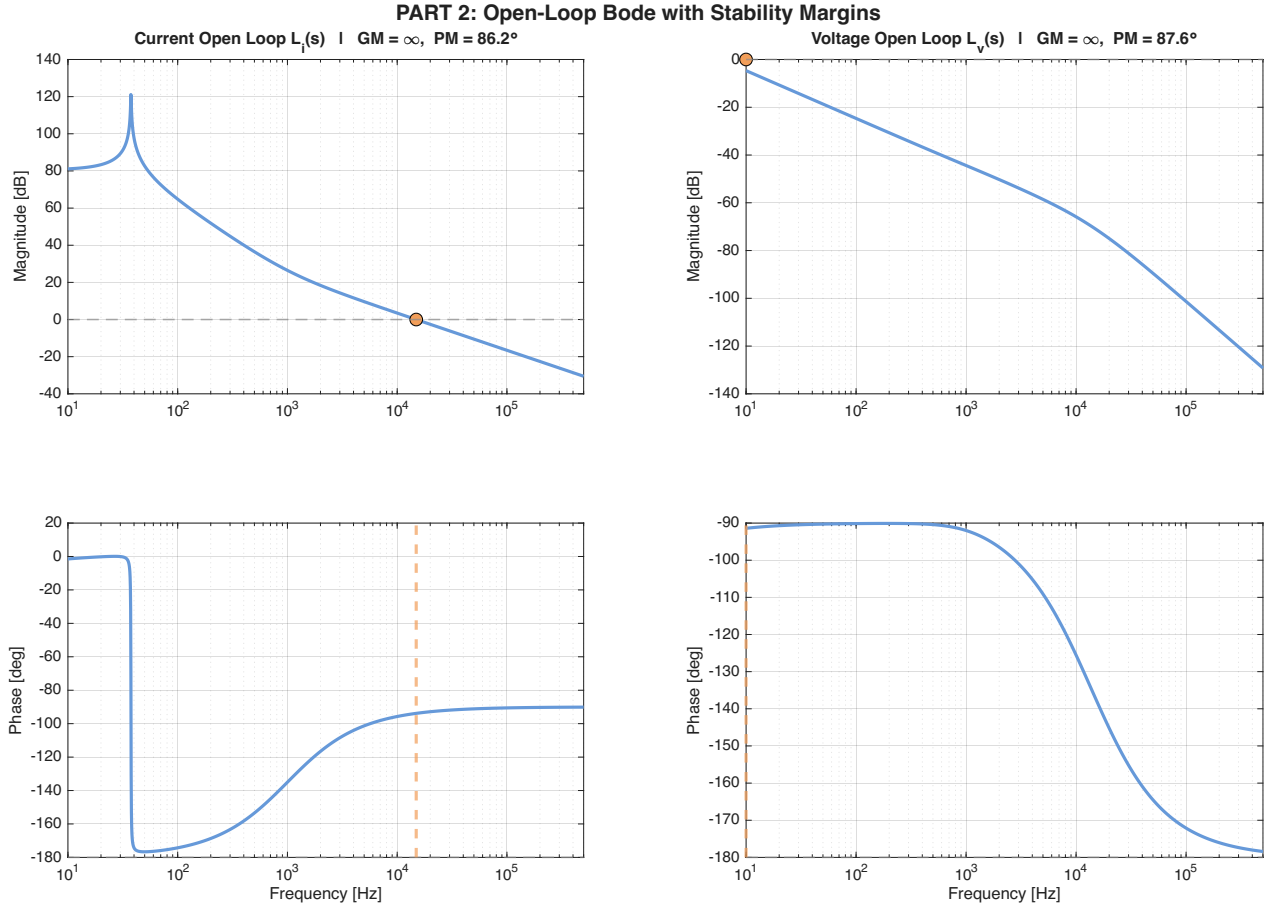


Figure 8: Open-loop Bode plots and stability margins of the implemented current and voltage loops. The current loop has $f_{c,i} \approx 14.89$ kHz and $PM_i \approx 86.2^\circ$. The voltage loop has $f_{c,v} \approx 5.85$ Hz and $PM_v \approx 87.6^\circ$.

6.3 Closed-loop Bode plots

The closed-loop Bode plots are shown in Figure 9. The current-loop bandwidth is higher than the original 5 kHz design hint because the final current PI gain is relatively high. The voltage-loop closed-loop bandwidth is also above the nominal target, but it remains much lower than 50 Hz. Therefore, the voltage loop is still sufficiently slow compared with the line frequency, which is important for avoiding distortion of the sinusoidal current reference.

PART 2: Closed-Loop Bode Plots

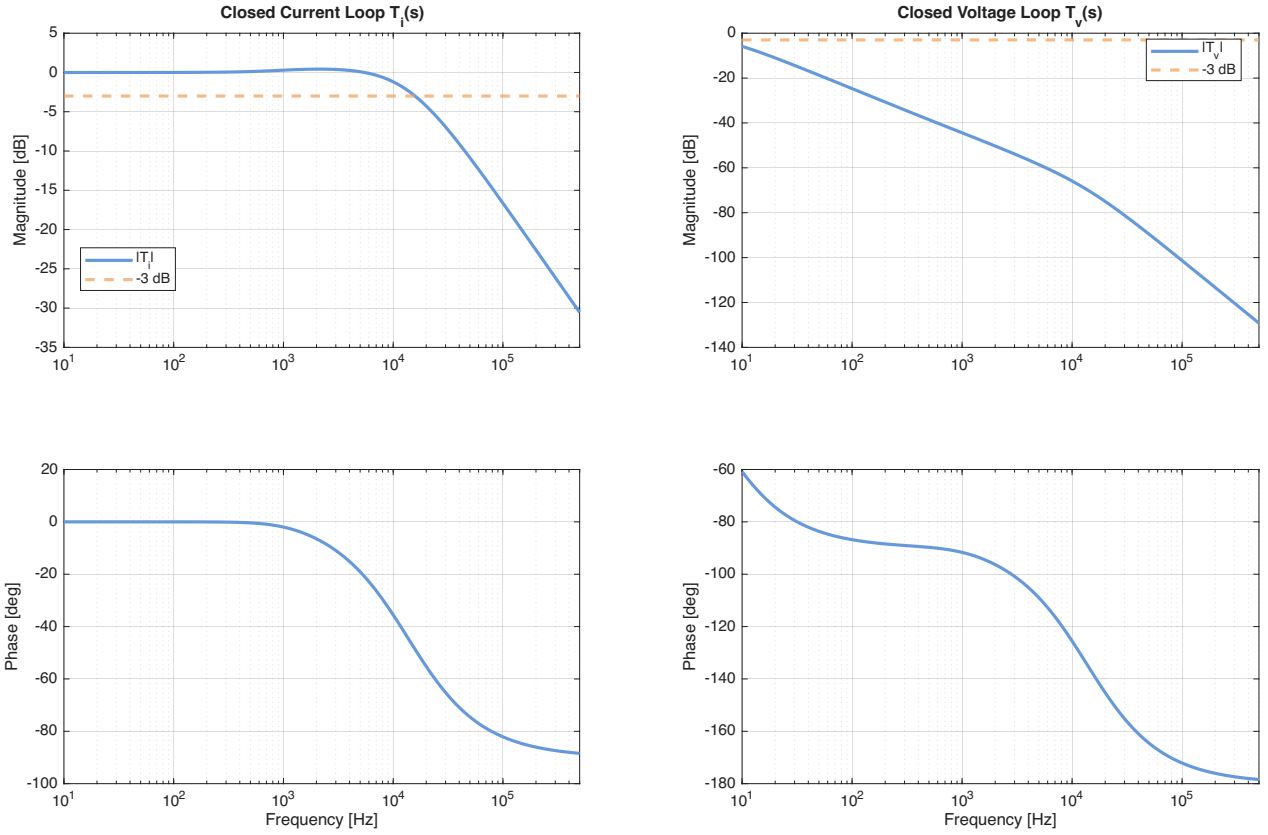


Figure 9: Closed-loop Bode plots of the implemented current loop $T_i(s)$ and voltage loop $T_v(s)$.

6.4 Summary

The final frequency-domain results are summarized in Table 5. Both phase margins are well above the usual 45° minimum. Therefore, the main issue is not instability. The main design compromise is that the final controller is faster than the original crossover targets, but this controller and the final passive values gave the best practical THD/PF result among the tested cases.

Table 5: Frequency-domain summary of the implemented compensated loops.

Quantity	Current loop	Voltage loop	Evaluation
Target crossover	5 kHz	2.5 Hz	Design hint
Actual crossover	14.89 kHz	5.85 Hz	Higher than target
Ratio to target	$\approx 3.0\times$	$\approx 2.3\times$	Accepted compromise
Phase margin	$\approx 86.2^\circ$	$\approx 87.6^\circ$	Stable
Gain margin	∞	∞	No finite crossing
Final decision	Selected based on best observed THD/PF, not only crossover matching		Practical design

7 PART (f): CLOSED-LOOP PERFORMANCE, PF AND THD EVALUATION

This section evaluates the final closed-loop switched simulation using $L = 12 \text{ mH}$, $C_o = 1500 \text{ }\mu\text{F}$, and the final selected PI parameters of Section 5. This part is used as the main reference case for comparing the later implementation stages in Parts (g) and (h). The goal is to verify output-voltage regulation, input-current shaping, power factor, and input-current THD.

7.1 Closed-loop waveforms

The final closed-loop waveforms are shown in Figure 10. The output voltage settles around the 400 V reference after the startup transient. The input voltage is sinusoidal, while the input current is shaped by the current loop and remains bounded around the line-voltage waveform. The inductor, diode, capacitor, and switch currents also remain within stable operating ranges.

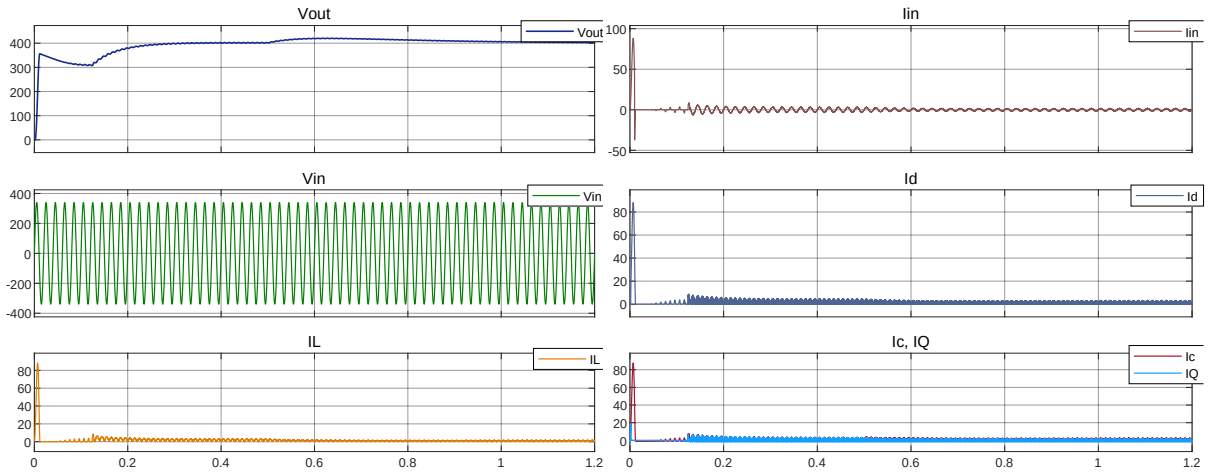


Figure 10: Final closed-loop simulation waveforms for Part (f) over the selected simulation interval.

7.2 Power factor and THD

The measured power factor was approximately

$$\text{PF} \approx 0.99, \quad (56)$$

which indicates that the input current is almost in phase with the input voltage. The THD display in the final model gave a representative value of approximately

$$\text{THD} = 0.0448 \approx 4.48\%, \quad (57)$$

which is below the 5% project target. The model screenshot showing the PF and THD measurement blocks is included in Figure 11.

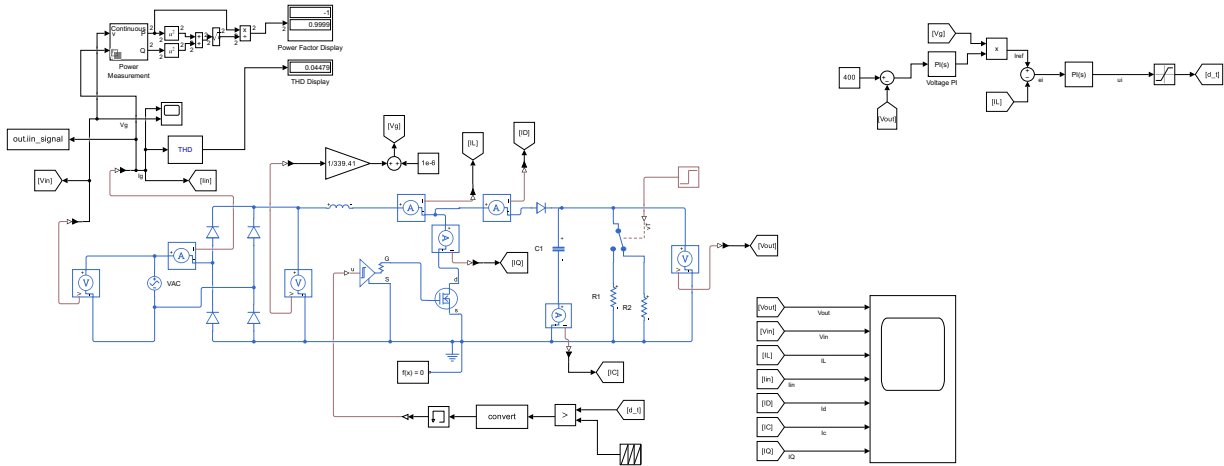


Figure 11: Final Simulink/PLECS model view with the THD and power-factor measurement displays. The shown representative THD value is 0.0448 and the PF is approximately 0.99.

During repeated simulation runs, the THD was usually kept below 5 %; however, short intervals or measurement-window changes sometimes produced temporary THD increases up to approximately 12 %. For this reason, the final result should be interpreted as acceptable in the selected steady-state window, but not perfectly robust against all simulation-window selections. The high current-loop bandwidth observed in Section 6 is one possible reason for this sensitivity. Nevertheless, among the tested PI and passive-value combinations, this final configuration gave the best overall compromise between regulation, PF and THD.

7.3 Result summary

Table 6: Summary of the final Part (f) closed-loop simulation results.

Parameter	Obtained result	Evaluation
Inductor and capacitor	$L = 12 \text{ mH}$, $C_o = 1500 \text{ }\mu\text{F}$	Final passive values
Output voltage	$\approx 400 \text{ V}$	Regulated
Power factor	≈ 0.99	Satisfactory
Representative THD	$\approx 4.48\%$	Below 5 % target
Observed THD spikes	Up to $\approx 12\%$	Intermittent/window dependent
PI gains	Best set found after trials	Practical optimum for final model

Overall, the closed-loop converter achieves the desired DC output voltage, a power factor close to unity, and a representative THD value below the 5 % limit. The remaining limitation is the occasional THD increase, which should be evaluated together with the fact that the final PI values were selected from practical simulation trials rather than only from ideal crossover-frequency matching.

8 PART (g): MODEL-IN-THE-LOOP IMPLEMENTATION AND SIMULATION OBSERVATIONS

In Part (g), the controller was checked in a Model-in-the-Loop style implementation using the same final passive components and PI values. The purpose of this stage is to verify that the controller structure can be implemented in a more digital/simulation-oriented form and to observe whether the main closed-loop behavior of Part (f) is preserved.

8.1 Simulation result

The available Part (g) waveform set is shown in Figure 12. The same main converter variables are monitored: output voltage, input voltage, inductor current, input current, diode current, capacitor current, and switch current. The output voltage remains close to the regulated operating point after the startup transient, and the current waveforms remain bounded.

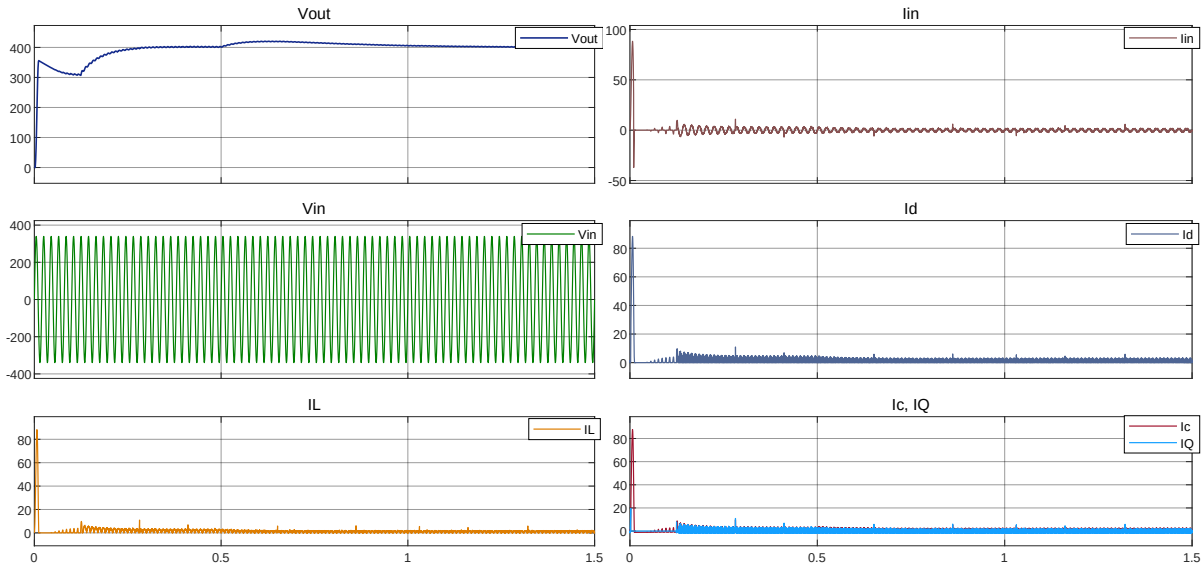


Figure 12: Model-in-the-Loop simulation waveforms obtained in Part (g) using the final passive values and final selected PI gains.

8.2 THD measurement model

For Part (g), the THD measurement circuit/model is shown in Figure 13. The displayed representative THD value is approximately $0.04922 \approx 4.92\%$, while the displayed power factor is approximately 0.9999. Therefore, Part (g) also satisfies the 5 % THD target in the selected steady-state measurement window.

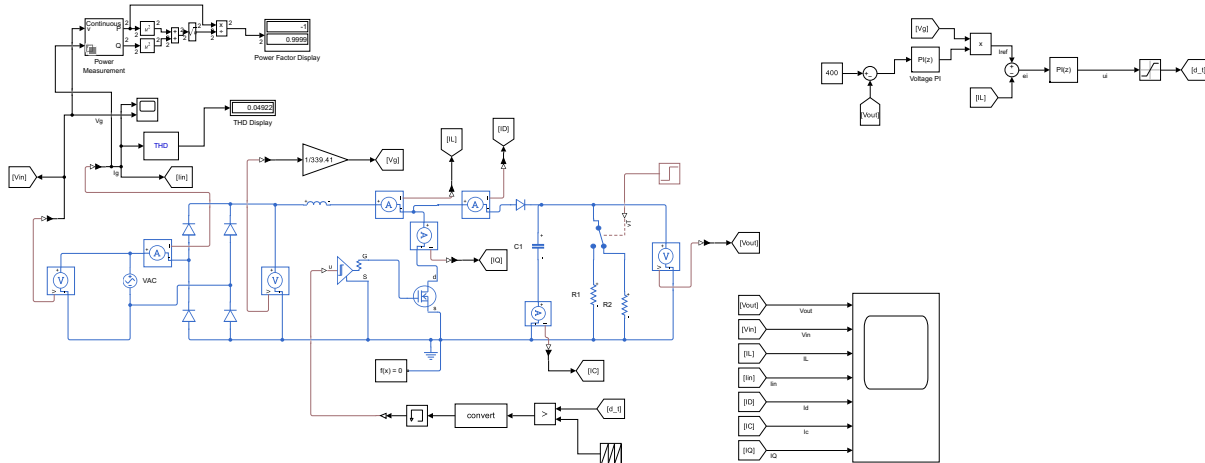


Figure 13: Model-in-the-Loop implementation view used in Part (g), including the THD and power-factor measurement blocks. The displayed representative THD is approximately 0.04922 and the power factor is approximately 0.9999.

8.3 Comparison with Part (f)

Part (f) represents the main closed-loop switched simulation reference, where the representative THD was approximately 4.48 % and the power factor was approximately 0.99. Compared with that reference case, Part (g) preserves the same general operating behavior: the output voltage stays close to 400 V, the input current is controlled, and the internal converter currents remain bounded. Therefore, the MIL implementation confirms that the selected controller can be transferred to the implementation-oriented model without losing basic stability.

The main difference is that the Part (g) waveforms are more affected by digital implementation details, switching events and solver interaction. The input-current waveform contains visible ripple and distortion, so the harmonic performance is expected to be more sensitive to the selected THD measurement window. However, the added THD/PF measurement model shows that a representative THD value of approximately 4.92% can still be obtained in Part (g). In other words, Part (g) is qualitatively consistent with Part (f), but it does not improve the THD robustness beyond the limitation already observed in Part (f).

Table 7: Comparison between Part (f) and Part (g).

Criterion	Part (f)	Part (g)
Implementation type	Closed-loop switched model	MIL-style implementation
Passive values	$L = 12 \text{ mH}$, $C_o = 1500 \text{ }\mu\text{F}$	Same values
PI gains	Final selected continuous values	Same selected values / discrete form
Output voltage	Around 400 V	Around 400 V
Input current	Shaped, with ripple	Shaped, but more implementation-sensitive
THD/PF behavior	THD $\approx 4.48 \%$, PF ≈ 0.99 in selected window	THD $\approx 4.92 \%$, PF ≈ 0.9999 in selected window; more window/solver sensitive

Thus, Part (g) confirms that the controller can be implemented in the model structure. It also shows a representative THD result below 5 %, similarly to Part (f). However, the final harmonic performance remains sensitive to the high current-loop bandwidth and simulation/measurement settings.

9 PART (h): SIL IMPLEMENTATION AND FINAL COMPARISON

In Part (h), the controller was evaluated in a Software-in-the-Loop (SIL) implementation. This stage is used as the final implementation-oriented verification after the closed-loop simulation in Part (f) and the MIL-style implementation in Part (g). The same final passive values and PI constants are used:

$$L = 12 \text{ mH}, \quad C_o = 1500 \text{ }\mu\text{F}, \quad (58)$$

$$K_{p,v} = 0.055, \quad K_{i,v} = 0.2, \quad K_{p,i} = 2.8, \quad K_{i,i} = 17600. \quad (59)$$

For the discrete implementation, the sample time is $T_s = 20 \text{ }\mu\text{s}$, which gives $K_{i,v,d} = 4.0 \times 10^{-6}$ and $K_{i,i,d} = 0.352$.

The Part (h) SIL implementation was carried out on a macOS system using the Xcode Clang compiler with a macOS x64 architecture. Therefore, if the same SIL setup is opened or executed on a Windows-based system, compiler-, path-, or toolchain-related errors may be observed due to differences between the macOS/Xcode-Clang environment and the Windows compiler environment.

9.1 SIL waveform result

The SIL waveform result is shown in Figure 14. The output voltage initially drops during the startup transient and then rises toward the 400 V operating point. Around the middle of the simulation, the output voltage exhibits a small overshoot above 400 V and then returns close to the reference. The input voltage remains sinusoidal. The input current, inductor current, diode current, capacitor current and switch current include noticeable startup peaks, but they remain bounded after the initial transient.

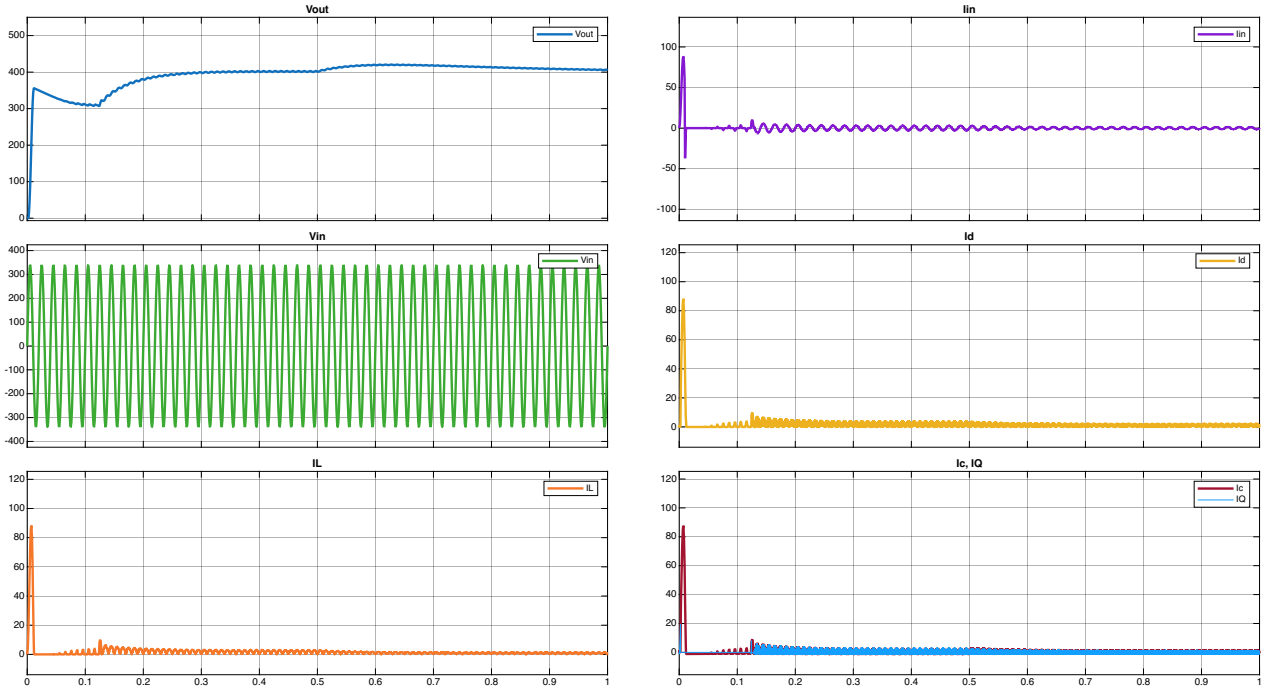


Figure 14: Software-in-the-Loop waveform result for Part (h). The monitored variables are output voltage, input voltage, inductor current, input current, diode current, capacitor current and switch current.

9.2 Simulation Data Inspector graph

In addition to the SIL waveform set, the Simulation Data Inspector graph is included in Figure 15. This graph provides a clearer view of the output-voltage and input-current behavior over the selected time interval. The output voltage drops at startup, recovers toward 400 V, shows a small overshoot above the reference, and then settles close to the desired operating point. The input current also exhibits the expected startup transient and then remains bounded around its steady-state behavior.

6/7/26, 9:36 PM

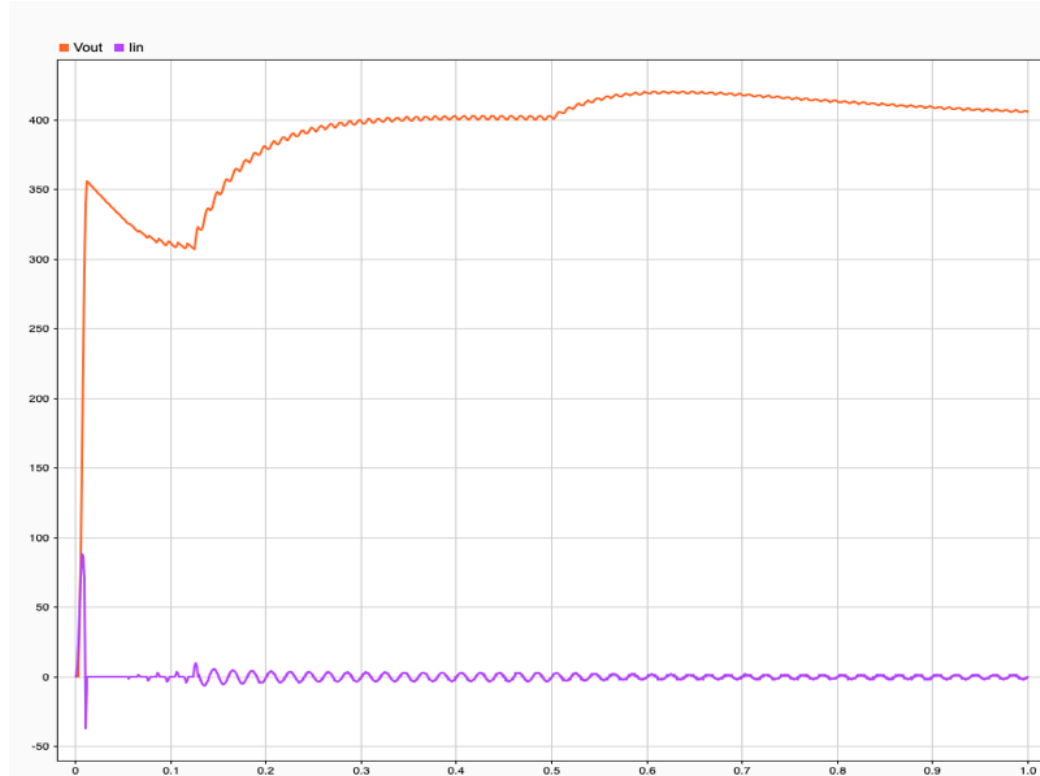
Untitled.html

Simulation Data Inspector: Inspect

Report Generated 07-Jun-2026 21:34:56

Run 3: PFC_Converter_Model_H

Name	Line
Vout	—
Iin	—



file:///Users/apple/Desktop/Untitled.html

1/1

Figure 15: Simulation Data Inspector graph for Part (h), showing the SIL result for V_{out} and I_{in} . This figure could not be exported directly as PDF from Data Inspector, so it was first saved as an HTML report and then printed/exported from the web view as a PDF. Under normal conditions, the result could be saved as data, but direct PDF export was not available in this case.

9.3 Comparison with Parts (f) and (g)

Compared with Part (f), the SIL result preserves the main closed-loop objective: the DC-link voltage returns close to the 400 V reference and the converter currents remain stable after the transient. However, Part (f) is still the cleanest reference case because it includes the direct PF/THD measurement result, where the representative THD is approximately 4.48 % and the power factor is approximately 0.99. In Part (h), the comparison is mainly waveform-based because the provided SIL graph and the Simulation Data Inspector graph show the time-domain behavior rather than a separate PF/THD display.

Compared with Part (g), Part (h) shows a similar implementation-oriented behavior. Both cases verify that the final controller can be executed outside the ideal continuous-controller reference model. The main difference is that the SIL waveform and Data Inspector graph contain stronger startup peaks and a more visible output-voltage transient. This is expected because the SIL implementation includes additional effects related to discrete execution, code generation and interface timing.

Table 8: Final comparison of Parts (f), (g), and (h).

Criterion	Part (f)	Part (g)	Part (h)
Implementation level	Closed-loop reference	MIL-style model	SIL implementation
Passive values	$L = 12 \text{ mH}$, $C_o = 1500 \text{ }\mu\text{F}$	Same	Same
PI values	Final selected values	Same / discrete form	Same / discrete form
Output voltage	Regulated near 400 V	Regulated near 400 V	Returns near 400 V after transient
Input/current behavior	Bounded, shaped current	Bounded but more ripple-sensitive	Bounded with stronger startup peaks
THD/PF information	THD $\approx 4.48 \%$, PF ≈ 0.99	Qualitative implementation check	Qualitative SIL waveform and Data Inspector check
Main conclusion	Best reference result	Controller transfer verified	Software-level execution verified; Data Inspector trend confirmed

Overall, Parts (g) and (h) support the result obtained in Part (f). The final controller and passive values achieve stable voltage regulation and acceptable current behavior in the implementation-oriented models. The main limitation remains the same: the current-loop crossover is higher than the original target, so the THD result is sensitive to the operating window and implementation details. Nevertheless, the final passive redesign from $L = 1 \text{ mH}$ to $L = 12 \text{ mH}$ and from $C_o = 4500 \text{ }\mu\text{F}$ to $C_o = 1500 \text{ }\mu\text{F}$ made it possible to reach healthy THD values while keeping the power factor around 0.99.

10 PART (i): PIL IMPLEMENTATION (CONCEPTUAL) AND MATLAB WORKFLOW

In Part (i), the final stage of the model-based design flow, the Processor-in-the-Loop (PIL) implementation, is addressed. The previous stages followed the standard verification chain: the controller was first validated as a continuous closed-loop reference in Part (f), then transferred to a Model-in-the-Loop (MIL) form in Part (g), and finally executed as generated code on the host machine in the Software-in-the-Loop (SIL) stage in Part (h). PIL is the next step in this chain, where the same generated controller code runs on the actual target microcontroller (MCU) instead of the host computer.

Because a physical C2000 microcontroller board and the associated Code Composer Studio toolchain were not available during this project, the PIL stage could not be executed on real hardware. Therefore, this part documents how PIL works, how the controller would have been deployed if the target MCU had been available, and the MATLAB/Simulink workflow that would have been used. The code listings below show the exact commands and configuration that would have driven the PIL run, so that the procedure is fully reproducible once the hardware is present.

10.1 What PIL is and how it works

PIL closes the gap between simulation and a real embedded implementation. In MIL and SIL the controller is still evaluated on the host: MIL runs the controller as a model, while SIL compiles the generated C code for the host CPU and runs it on the host. PIL instead cross-compiles the generated code for the instruction set of the target processor, downloads the executable to the MCU, and runs the controller *on the MCU* while the rest of the plant (the boost converter, the line source and the load) continues to run as a model on the host.

The two sides communicate over a link such as a serial/UART or JTAG connection. At every controller sample instant the host sends the controller block inputs (here the sensed output voltage V_{out} and the sensed inductor/input current) to the target, the target executes the generated control code, and the resulting output (the duty cycle / PWM reference) is sent back to the host and injected into the plant model. As a result, PIL verifies three things that SIL cannot fully guarantee:

- **Target numerics:** the controller now runs with the real data types of the MCU (for a C2000 device, single-precision floating point on the hardware FPU), so any deviation caused by finite word length is exposed.
- **Bit-accurate behavior:** the result is numerically identical to what the deployed firmware would produce, because it is the same cross-compiled code.
- **Execution timing:** code-execution profiling reports the worst-case execution time of the control loop, which must remain below the sample period $T_s = 20 \mu s$ for the controller to be feasible on the chosen MCU.

10.2 Intended target and how it would have been implemented

For an average-current-mode PFC boost converter, the natural target is a Texas Instruments C2000 real-time MCU, for example the LAUNCHXL-F28379D (Delfino F2837xD) board. This family is the standard choice for digital power-electronics control and is consistent with the average-current PFC reference used in this project [4], which was implemented on the same C2000 lineage. The required software components are Embedded Coder, the C2000 Microcontroller Blockset, and the Code Composer Studio compiler toolchain.

The discrete controller subsystem prepared for the SIL stage would be reused directly, with the same sample time $T_s = 20\ \mu\text{s}$ and the same discrete PI gains. The implementation steps would be:

1. Select the C2000 target device under *Configuration Parameters* $\rightarrow$ *Hardware Implementation*.
2. Set Embedded Coder (`ert.tlc`) as the system target file and configure the PIL connectivity (serial connection to the board).
3. Right-click the discrete controller subsystem and choose *SIL/PIL* $\rightarrow$ *Run as Processor-in-the-Loop*, or drive the run from the SIL/PIL Manager.
4. Embedded Coder generates the C code, cross-compiles it for the C2000, and downloads it to the MCU automatically.
5. Run the closed-loop simulation; the host plant model and the MCU exchange the controller I/O at every sample.
6. Compare the PIL output (V_{out} , I_{in} , duty cycle) against the MIL/SIL runs in the Simulation Data Inspector, and read the execution-profiling report to confirm timing feasibility.

10.3 MATLAB/Simulink workflow examples

The following listings show the commands that would have been used to configure and run the PIL stage. They assume that the discrete controller subsystem from Part (h) is reused inside a model named `PFC_Controller_PIL`.

(1) Target and solver configuration. The model is pointed at the C2000 device, Embedded Coder is selected, and a fixed-step discrete solver matching the controller sample time is enforced. Code-execution profiling is turned on so that the loop timing can be measured.

```

1 mdl = 'PFC_Controller_PIL';
2 load_system(mdl);
3
4 % --- Target processor: TI C2000 (Delfino F2837xD LaunchPad) ---
5 set_param(mdl,'HardwareBoard', ...
6   'TI Delfino F2837xD LaunchPad (LAUNCHXL-F28379D)');
7 set_param(mdl,'SystemTargetFile','ert.tlc'); % Embedded Coder (ERT)
8 set_param(mdl,'TargetLang','C');
9
10 % --- Fixed-step discrete solver matching Ts = 20 us ---
11 set_param(mdl,'SolverType','Fixed-step');
12 set_param(mdl,'Solver','FixedStepDiscrete');
13 set_param(mdl,'FixedStep','20e-6');
14
15 % --- Enable code-execution profiling for the PIL run ---
16 set_param(mdl,'CodeExecutionProfiling','on');
17 set_param(mdl,'CodeProfilingInstrumentation','coarse');
18 set_param(mdl,'DefaultParameterBehavior','Inlined');
```

Listing 1: Configuring the model for the C2000 target and enabling PIL profiling.

(2) Running the controller subsystem as PIL. The controller subsystem is switched into PIL simulation mode, so that during the run it executes on the MCU while the plant runs on the host.

```
1 % Path of the discrete average-current controller subsystem
2 ctrl = 'PFC_Controller_PIL/Average_Current_Controller';
3
4 origMode = get_param(ctrl,'SimulationMode');           % store original
5 set_param(ctrl,'SimulationMode', ...
6     'Processor-in-the-Loop (PIL)');
7
8 % Run the closed loop; controller code runs on the C2000 target
9 out = sim('PFC_Controller_PIL','StopTime','1.5');
10
11 set_param(ctrl,'SimulationMode',origMode);           % restore
```

Listing 2: Running the controller subsystem in Processor-in-the-Loop mode.

(3) Comparing PIL against MIL/SIL in the Data Inspector. Each run is logged automatically. The PIL run is then compared against the earlier MIL/SIL run with a tolerance, which is the quantitative way to confirm that the on-target result matches the verified model.

```
1 runIDs = Simulink.sdi.getAllRunIDs;
2 refRun = Simulink.sdi.getRun(runIDs(end-1));           % previous MIL/SIL run
3 pilRun = Simulink.sdi.getRun(runIDs(end));             % current PIL run
4
5 % Compare runs with an absolute tolerance (e.g. 0.5 V / 0.5 A)
6 diffResult = Simulink.sdi.compareRuns( ...
7     refRun.ID, pilRun.ID, 'absTol', 0.5);
8
9 disp(diffResult.Summary); % within-tolerance vs out-of-tolerance signals
```

Listing 3: Comparing the PIL run against a reference run in the Simulation Data Inspector.

(4) Reading the execution-timing report. Finally, the profiling result is used to check that the controller fits inside the $20\mu\text{s}$ sample period. If the worst-case execution time exceeded the budget, the controller would not be realizable on this MCU at this sample rate.

```
1 profile = out.executionProfile; % coder.profile.ExecutionTime
2 report(profile);                % opens the timing report
3
4 tMax = max([profile.Sections.MaximumExecutionTimeInSeconds]);
5 fprintf('Worst-case controller exec time: %.2f us (budget = %.1f us)\n', ...
6     tMax*1e6, 20);
```

Listing 4: Retrieving the code-execution profile and checking timing feasibility.

10.4 Expected outcome and limitation

If the hardware had been available, the PIL waveforms for V_{out} and I_{in} would be expected to overlay the MIL/SIL results from Parts (g) and (h) almost exactly, with only small deviations attributable to single-precision floating-point arithmetic on the C2000 FPU compared with the double-precision host computation. The execution-profiling report would then confirm whether the average-current controller and the outer voltage loop together fit within the $20\mu\text{s}$ sample period.

The limitation of this part is therefore purely practical: the verification chain MIL $\rightarrow$ SIL was completed successfully in Parts (g) and (h), but the final PIL link could not be closed on real hardware because the target MCU and its toolchain were not accessible. The workflow above is provided so

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Students : Enes Aytuğ KIR & Ufuk Barın



that the PIL stage can be executed and verified directly once a C2000 board is available, without any change to the controller design or the discrete PI gains.

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